

Facebook Ad Analysis Dashboard

This project is an end-to-end analysis and visualization of Facebook ad campaign performance using real-world data. It focuses on identifying top-performing ads, understanding user engagement, and analyzing demographic insights through a Power BI dashboard.

Dataset

- Sourced from Kaggle
 - Contains detailed Facebook ad data including impressions, clicks, spend, age, gender, and conversion metrics
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Data Preprocessing

- Used SQL to clean and prepare the data for analysis
 - Handled missing values and transformed fields as needed
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Metrics and DAX Calculations

Custom DAX measures were created in Power BI to evaluate key performance indicators, including:

- Click-Through Rate (CTR): $\text{Clicks} / \text{Impressions}$
 - Conversion Rate: $\text{Approved Conversions} / \text{Clicks}$
 - Cost per Join: $\text{Spend} / \text{User Joins}$ (optional)
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Dashboard Features

The dashboard was created in Power BI using a custom background designed in PowerPoint. Key visualizations include:

- KPI cards for Total Spend, Total Clicks, CTR, and Conversion Rate
 - Bar and donut charts showing performance by age and gender
 - Top 3 ads by number of user signups
 - Trend chart of ad spend over time, segmented by gender
 - Scatter plot of ad spend vs. number of signups with click-based bubble sizing
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Tools Used

- SQL for data cleaning
 - Power BI for dashboard creation
 - DAX for calculated measures
 - PowerPoint for background design
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Key Insights

- Higher click volumes generally correlated with a higher number of user accounts created
- Demographic breakdowns showed varying engagement across age and gender
- The top-performing ads had strong cost efficiency in driving user acquisition